

THE SUNSPOT AND AURORAL ACTIVITY CYCLE DERIVED FROM KOREAN HISTORICAL RECORDS OF THE 11th–18th CENTURY

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Abstract. Historical records of sunspots and aurorae are valuable information to examine variations of solar activity and the terrestrial climate on a long-term scale. We have collected the historical records of Korea during the 11th–18th century. Through a power-spectrum analysis of these data, we have found solar activity cycles, which coincide with the Schwabe cycle and the Gleissberg cycle on short and long-term periods, respectively.

1. Introduction

The variations of solar activity are observed directly from sunspots or indirectly detected through proxy data of solar wind, polar aurora, or cosmogenic isotope concentration as ^{10}Be in polar ice and ^{14}C in tree rings. In this study, we have examined the historical records of sunspots and aurorae for the research of long-term solar variation. The use of historical data sets to investigate solar and solar-terrestrial variability was demonstrated dramatically by J. A. Eddy when he described the Maunder Minimum (Feynman, 1988). Although the historical data such as sunspot and auroral observations are fragmentary and strongly depend on the frequency of observation, they are still important evidence and information of solar activity on a long-term scale. It is known that there is a strong correlation between the long-term occurrence of sunspot numbers and the frequency of aurorae (Silverman, 1992, 1998; Usoskin and Mursula, 2003). Auroral activity shows 11-year and 22-year variations (Attollini *et al.*, 1988) as well as the well-established strong 88-year cycle (Gleissberg, 1955; Siscoe, 1980; Feynman *et al.*, 1984, 1989). In this paper, we discuss the features of long-term variations seen in the distribution of auroral records, and solar activity cycles derived from the power spectrum analysis of historical data, related to the 11-year and 88-year cycles.

2. The Historical Records of Sunspots and Aurorae

A lot of the sunspot and auroral data are recorded in the historical annals and documents of Korea. Although some auroral catalogues containing Korean records

were published (see, e.g., Liu, 1988; Yau, Stephenson, and Willis, 1995), lots of records are missing in these lists, which referred only to limited documents. In this work, we have collected and examined the historical records of Korea from the following sources.

- 1) *Koryo-Sa*: the Annals of Koryo Dynasty (918–1391).
- 2) *Choson Wangjo Sillok*: the Annals of Choson Dynasty (1392–1910).
- 3) *Jeungbo Munheon Bigo*: an encyclopedia of 250 volumes, which was published in 1770 and revised later in 1908.
- 4) *Seungjeongweon Ilgi*: daily journals written by official-scholars in the office of the royal secretary during Choson Dynasty.
- 5) *Daedong Yaseung*: unofficial history book of 71 volumes, during Choson Dynasty.

From the above sources, we have collected 60 sunspot and 788 auroral records during the 11th–18th centuries. These are listed in Tables I and II for sunspots and aurorae, respectively. Originally recorded dates by luni-solar calendar were converted to Gregorian (before 26 Oct. 1582) and Julian calendar in tables. In these records, we know that auroral observations are recorded much more frequently than sunspots. In case of Choson Dynasty, having 17 sunspot and 542 auroral records, such a difference is much larger than Koryo Dynasty. According to the size of sunspot or intensity of auroral activity, the shapes of sunspots were described as plum, egg, peach, pear and knife etc., and the color and shape of aurorae, as a flame, fire, glow, rainbow, silk, dragon and snake etc. Observed auroral color in the records was absolutely dominant as red, which means observation at low latitude.

3. The Distribution of Auroral Records

The number of historical reports of auroral nights or sunspot days indicates the existence of episodes such as the Maunder Minimum, and implications for secular solar behavior (Eddy, 1983). Therefore, we have examined the secular variation of solar activity, through the number distributions of many auroral records during the 11th–18th centuries (see Figure 1). During this period, the existence of periods now identified as the Maunder Minimum (1645–1715), the Spörer Minimum (1420–1530), Wolf Minimum (1280–1340), and Oort Minimum (1010–1050) were established by measurement of radiocarbon data in tree-rings (Stuiver and Quay, 1980; Stuiver and Grootes, 1980; Eddy, 1983). In spite of the distribution of non-linear auroral data, Figure 1 shows that Great Minima such as the Oort, Wolf, Spörer and Maunder Minima clearly appeared in the times of low auroral activities. It also shows that high-intensity auroral activities existed after the Spörer Minimum and before the Maunder Minimum.

On the other hand, Schröder (1994) reported that auroral activity during the period 1450–1550 was normal, using auroral data from Central Europe (Austria, Czechoslovakia, Germany, Hungary, Switzerland). However, his suggestion

TABLE I
Sunspot records.

Year	Month	Day	Description	Year	Month	Day	Description
1105	2	7	Black light	1371	1	2	Black spot
1151	3	21	Black spot as large as hen's egg	1371	11	21	Black spot
1151	3	31	Black spot as large as hen's egg	1372	5	8	Black spot
1152	4	1	Black spot as large as hen's egg	1373	4	26	Black spot
1160	9	29	Black spot	1373	4	27	Black spot
1171	10	20	Black spot as large as peach	1373	10	23	Black spot
1171	11	16	Black spot as large as peach	1375	3	20	Black spot
1183	12	4	Black spot	1375	3	21	Black spot
1183	12	5	Black spot	1381	3	23	Black spot
1185	2	11	Black spot as large as pear	1382	3	9	Black spot as large as hen's egg
1185	3	27	Black spot as large as pear	1382	3	10	Black spot as large as hen's egg
1185	4	18	Black spot	1382	3	11	Black spot as large as hen's egg
1185	4	19	Black spot	1387	4	15	Black spot
1185	11	14	Black spot	1402	11	15	Black spot
1200	9	19	Black spot as large as plum	1520	3	9	Black gas
1201	4	6	Black spot as large as plum	1556	4	17	Black spot as large as hen's egg
1202	8	23	Black spot as large as pear	1603	4	16	Black spot as like big coin
1204	2	3	Black spot as large as plum	1604	10	24	Black spot as large as bird's egg
1204	2	4	Black spot as large as plum	1604	10	25	Black spot as large as hen's egg
1204	2	5	Black spot as large as plum	1608	5	10	Black spot as large as pear
1258	9	15	Black spot as large as hen's egg	1608	5	17	Black spot as large as pear
1258	9	16	Black spot like a doll	1648	1	16	Black spot
1278	8	31	Black spot as large as hen's egg	1660	5	22	Black gas
1356	4	4	Black spot	1720	5	8	Black gas
1356	4	5	Black spot	1720	6	1	Black gas
1361	3	16	Black spot	1726	10	21	Black gas
1361	3	17	Black spot	1726	10	22	Black gas
1361	3	18	Black spot	1743	10	19	Black gas
1361	3	19	Black spot	1743	10	20	Black gas
1362	10	5	Black spot	1743	10	21	Black gas

contradicts what we can see in Figure 1. It seems that he could not detect the real behavior of solar activity, because of insufficient auroral data from Central Europe. Jiang and Xu (1986), who examined the duration of the Spörer Minimum, said that the Spörer Minimum is a real feature of solar activity in the 15th century. Eddy (1976) also determined the period 1400–1510 as the real extent of the Spörer Minimum. Further, more research is needed in the future on the Spörer Minimum and the Maunder Minimum.

TABLE II
Auroral records.

Year	Month	Day	Color	Year	Month	Day	Color	Year	Month	Day	Color
992	12		R	1156	4	25	R	1253	8	11	R
1012	6	12	R	1174	1	6	R	1253	8	29	Y-R
1014	4	6	R	1175	11	13	R	1254	10	25	R
1017	1	27	R	1176	3	13	R	1255	10	2	R
1017	3	4	R	1176	3	14	R	1256	5	28	R
1017	12	15	R	1176	3	19	R	1257	3	14	R
1019	3	6	R	1176	9	13	R	1257	6	26	R
1024	12		R	1176	10	10	R	1257	7	22	R
1028	9	25	R	1177	2	19	R	1257	9	1	Y-R
1052	7	18	W, B-V	1177	3	23	R	1258	1	14	Y-R
1073	1	12	W-R	1177	8	2	R	1258	2	8	Y-R
1073	10	13	R	1177	12	28	R	1259	1	26	R
1073	12	28	B-R	1178	4	7	R	1259	7	17	R
1088	8	14	R	1178	4	15	R	1260	8	7	R
1090	2	25	R-W	1178	11	12	R	1260	10	30	R
1101	1	31	R	1178	11	15	R	1260	12	4	R
1104	2	7	R	1178	12	11	R	1260	12	26	R
1104	2	21	R	1178	12	14	R	1261	1	29	R
1105	1	19	R-W	1179	3	12	R	1262	10	10	R
1105	2	17	R	1180	3	28	R	1262	12	8	R
1105	2	24	Y-R	1180	8	23	R	1264	2	16	R
1108	2	3	B-W	1181	3	13	W, R	1268	12	19	R
1109	5	13	R, Y-BK	1181	9	4	R	1273	1	21	R
1113	3	11	R	1185	3	26	R	1274	1	26	R
1114	4	2	R	1187	10	1	R, Rainbow	1275	1	3	R-BK
1115	3	21	R	1187	11	18	R	1275	1	21	R
1116	10	10	R	1187	11	22	R	1276	12	2	R-W
1116	10	17	R	1188	10	29	R	1277	3	30	Rainbow
1117	12	16	R	1188	10	30	R	1277	4	29	R-W
1118	3	27	R	1188	11	1	R	1278	3	23	R
1121	2	23	R	1192	12	29	R	1278	3	26	R
1121	4	14	R	1195	4	8	R	1279	10	25	V
1122	4	21	B, R	1196	8	26	R	1282	2	10	R
1122	5	15	Rainbow	1196	11	9	R	1287	3	8	R
1123	3	21	W-R	1201	12	24	R	1288	11	13	R
1126	6	30	R	1217	4	12	R	1292	12	31	R
1126	8	4	R	1217	4	18	R	1294	2	1	R
1127	10	17	R	1217	4	21	R	1294	7	20	V, R

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TABLE II

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Year	Month	Day	Color	Year	Month	Day	Color	Year	Month	Day	Color
1127	10	18	R	1217	7	17	V	1296	2	3	R
1127	10	19	R	1220	3	23	R	1296	2	29	R-W
1127	10	20	R	1220	3	24	R	1304	1	28	R
1128	2	28	R-W	1220	3	25	R	1307	3	27	R
1128	7	24	Y-R	1221	9	12	R	1312	2	11	R
1128	10	20	R, BK	1222	5	4	R	1314	3	9	R
1128	12	13	R	1222	6	8	R	1316	3	24	R
1129	1	10	R	1222	9	1	R	1316	10	12	R
1129	10	25	R	1224	10	15	R	1320	2	1	R
1129	12	27	R	1225	8	16	R	1320	11	17	R
1130	3	31	R	1226	Fall		Y-R	1321	1	30	R-W
1130	6	19	R-BK	1227	4	4	R	1321	4	11	R
1130	9	29	R	1227	7	20	R, Rainbow	1323	8	19	Rainbow
1131	1	31	R	1227	8	27	R	1324	3	21	R
1131	2	5	R	1228	3	30	R	1324	4	20	R
1137	2	17	R	1229	9	9	R	1347	8	12	R
1138	8	28	R	1249	3	17	R	1351	12	23	V
1138	9	3	R	1250	11	25	R	1353	3	2	V
1138	10	6	R	1250	12	28	R	1357	11	19	V
1138	12	2	R	1252	1	2	R	1358	2	28	V
1141	11	22	R	1253	4	1	Y-R	1358	4	11	R
1141	12	23	R	1253	4	19	R	1359	2	20	V
1361	2	10	R	1381	2	27	R	1524	2	1	R
1362	12	19	V	1383	12	29	R	1524	3	26	R
1364	2	26	R	1385	3	7	R	1524	9	23	R
1364	3	1	Rainbow	1390	3	11	R	1524	11	19	R
1365	3	14	R	1390	4	29	W-R	1525	1	15	R
1365	3	17	R	1391	2	6	R	1525	2	13	R
1365	3	18	R	1397	4	1	R	1525	3	18	R
1365	3	19	R	1397	4	7	R-W	1525	3	28	R
1365	8	15	R	1397	12	22	R	1525	3	29	R
1366	11	30	R	1400	7	23	R	1525	4	6	R
1367	2	7	R	1401	2	5	R	1525	4	10	R
1367	2	28	R	1401	2	7	R	1525	4	15	W
1367	3	1	R	1402	1	7	R	1525	4	18	R
1367	3	2	R	1403	2	16	R-W	1525	4	22	R
1367	3	4	R	1403	3	8	R	1525	5	1	R
1367	3	6	R	1403	12	1	R-B	1525	5	9	R

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TABLE II

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Year	Month	Day	Color	Year	Month	Day	Color	Year	Month	Day	Color
1367	3	24	R	1403	12	14	R	1525	11	13	R
1367	6	13	R	1405	1	10	R	1526	3	21	R
1367	6	14	R	1405	1	21	R	1526	4	5	R
1367	11	21	R	1405	1	22	R	1526	4	8	R
1367	12	10	R	1405	1	23	R	1526	4	10	R
1367	12	11	R	1405	1	26	R	1526	4	30	R
1368	2	21	R	1405	3	11	R	1526	6	20	R
1368	3	10	R	1406	1	11	R	1526	6	30	R
1368	3	12	R	1406	1	16	R	1526	7	18	R
1368	3	16	R	1411	3	3	W-R	1526	8	5	R
1368	3	19	R	1467	8	19	R	1526	8	6	R
1368	3	20	R	1467	8	20	R	1526	8	7	R
1368	3	21	R	1467	8	21	R	1526	9	15	R
1368	3	22	R	1499	8	6		1526	9	24	R
1368	3	23	R	1507	1	24	R	1526	12	7	R
1369	12	15	R	1508	4	5	R	1526	12	14	R
1370	2	11	V	1509	12	17	R	1527	3	1	R
1370	3	17	R	1512	1	17	R-W	1528	3	17	R
1370	9	4	R	1512	3	21	R	1528	3	19	R
1370	11	24	R	1515	3	15	R	1528	3	31	W-R
1371	2	4	R	1515	3	18	R	1528	4	14	R
1371	10	19	R	1515	3	20	R	1528	6	15	R
1372	4	12	R	1515	3	21	R	1528	6	16	R
1373	1	11	R	1515	4	7	R	1529	3	8	R
1373	1	20	R-W	1515	4	8	R	1529	6	8	R
1373	1	27	R	1515	4	9	R	1530	2	21	R
1373	2	1	R	1515	4	13	R	1530	2	22	R
1373	2	16	R	1515	4	19	R	1531	2	24	R
1373	2	26	R	1517	2	27	R	1531	4	13	R
1373	3	3	R	1517	3	19	R	1531	5	23	R
1373	4	11	R	1517	3	20	R	1531	6	10	R
1373	9	8	R	1518	5	13	R	1531	7	19	R
1374	3	7	R	1519	7	2	R	1531	8	6	R
1374	3	10	R	1520	4	10	R	1532	3	12	R
1374	4	14	R	1520	4	12	R	1532	3	13	R
1374	4	16	R	1520	4	13	R	1532	4	3	R
1377	10	11	R	1522	3	20	R	1532	8	20	R
1379	1	11	R	1522	3	25	R	1532	9	26	R

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TABLE II
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Year	Month	Day	Color	Year	Month	Day	Color	Year	Month	Day	Color
1379	2	9	R	1522	3	31	R	1533	1	28	R
1379	3	15	R	1522	9	16	R	1533	2	2	R-W
1379	3	23	R	1522	11	20	R	1533	2	19	R
1380	4	2	R	1523	4	14	R	1533	2	21	R
1381	1	12	R	1523	11	9	R	1533	2	22	R
1381	1	31	R	1523	11	19	R	1533	3	26	R
1533	3	27	R	1538	3	5	R	1551	7	1	Y-R
1533	4	29	R	1538	6	1	R	1551	11	27	R
1533	4	30	R	1538	6	2	R	1551	12	1	R
1533	5	1	R	1538	6	24	R	1552	2	16	R
1533	5	2	R	1538	10	20	R	1552	2	27	R
1533	5	23	R	1538	10	21	R	1552	3	16	R
1533	7	25	R	1538	12	3	R	1552	3	30	R
1533	8	30	R	1538	12	4	R	1552	4	19	R
1533	10	24	B-Y-W	1539	1	8	R	1552	5	1	R
1534	1	13	R	1539	1	9	R	1552	5	28	R
1534	1	23	R	1539	1	10	R-W	1552	10	29	R
1534	4	14	R	1539	1	18	R	1552	11	1	Y-V
1534	4	16	R	1539	2	5	R	1553	1	15	R
1534	4	17	R	1539	3	21	R	1553	2	6	R
1534	4	23	R	1539	4	8	R	1553	2	7	R
1534	5	1	R	1539	4	9	R	1553	2	8	R
1534	5	20	R	1539	5	14	R	1553	3	10	R
1534	6	12	R	1539	5	29	R	1553	3	16	R
1534	6	13	R	1542	2	17	R	1553	3	18	R
1534	6	14	R	1542	3	12	R	1553	7	13	R
1534	9	2	R	1542	4	21	R	1553	8	8	R
1534	9	6	R	1542	6	8	R	1554	3	4	R
1535	1	4	R	1542	6	13	R	1554	3	9	R
1535	1	7	R	1542	6	21	R	1554	4	5	R
1535	1	8	R	1542	8	13	R	1554	4	21	R
1535	1	30	R	1542	10	26	R	1554	5	5	R
1535	3	6	R	1543	1	26	R	1554	6	10	R
1535	3	31	R	1543	2	5	R	1554	6	29	R
1535	4	2	R	1543	2	22	R	1554	7	7	R
1535	4	4	R	1543	2	27	R	1554	7	8	R
1535	4	29	R	1543	3	29	R	1554	8	7	R

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TABLE II
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Year	Month	Day	Color	Year	Month	Day	Color	Year	Month	Day	Color
1535	4	30	R	1543	3	30	R	1554	10	9	R
1535	5	6	R	1543	5	10	R	1554	12	16	R
1535	5	7	R	1543	5	23	R	1555	1	18	R
1535	5	30	R	1543	8	6	R	1555	2	12	R
1535	6	25	R	1544	3	29	R	1555	2	22	R
1535	6	30	R	1544	6	25	R	1555	2	27	R
1535	7	1	R	1544	7	18	B-W	1555	3	14	R
1535	7	28	R	1544	11	16	R	1555	3	16	R
1535	11	14	R	1545	4	1	R	1555	3	17	R
1535	11	30	R	1545	6	7	R	1555	3	22	R
1535	12	1	R	1546	1	4	R	1555	3	28	R
1535	12	22	R	1546	3	9	R	1555	4	19	R
1535	12	31	R	1546	5	9	R	1555	6	8	R
1536	2	17	R	1546	6	3	R	1555	6	13	R
1536	2	23	R	1546	7	14	Rainbow	1555	6	21	R
1536	3	7	R	1547	7	10	R	1555	7	13	R
1536	3	10	R	1548	1	1	R	1555	12	9	R
1536	3	12	R	1548	5	10	R	1556	1	7	R
1536	3	13	R	1548	7	3	R	1556	1	15	R
1536	3	17	R	1548	12	9	R	1556	3	2	R
1536	5	17	R	1549	1	24	R	1556	3	5	R
1536	8	19	R	1549	12	20	R	1556	3	8	R
1537	3	7	R	1550	5	14	R	1556	3	10	R
1537	6	2	R	1550	7	12	R	1556	3	11	R
1537	6	13	R	1551	3	15	R	1556	4	5	R
1537	6	13	R	1551	3	27	R	1556	4	7	R
1537	12	5	R	1551	4	1	R	1556	4	8	R
1538	1	20	R	1551	4	11	R	1556	4	13	R
1538	3	4	R	1551	5	27	R	1556	5	3	R
1556	6	4	R	1566	12	9	R	1624	2	26	R
1556	6	6	R	1567	5	7	R	1624	3	21	R
1556	6	10	R	1567	5	11	R	1624	4	18	R
1556	6	14	R	1588	7	24	R	1624	4	19	R
1556	6	19	R	1591	8		R	1624	4	21	R
1556	7	8	R	1593	11	27	R	1624	6	9	B-R
1556	8	11	R	1594	1	5	R	1624	7	11	R
1556	10	30	R	1594	1	13	R	1624	7	12	R
1556	11	11	R	1594	2	11	R	1624	12	31	R

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TABLE II

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Year	Month	Day	Color	Year	Month	Day	Color	Year	Month	Day	Color
1557	2	20	R	1594	3	20	R	1625	2	6	R
1557	3	25	R	1595	3	20	R	1625	2	9	R
1557	4	28	R	1595	4	30	R	1625	3	2	R
1557	6	4	R	1596	7	16	V	1625	3	6	R
1557	12	2	R	1596	10	31	B-R	1625	3	11	R
1557	12	12	B-W	1596	12	26	R	1625	3	15	R
1558	3	6	R	1597	2	5	R	1625	3	31	R
1558	3	16	R	1597	2	17	R	1625	4	7	R
1558	3	17	R	1597	2	18	R	1625	4	8	R
1558	4	16	R	1597	2	20	R	1625	4	11	R
1558	4	21	R	1597	2	22	R	1625	8	4	R
1558	5	4	R	1599	3	23	R	1625	8	13	B-W
1558	6	4	R	1599	3	27	R	1625	8	28	R
1558	11	6	R	1601	1	28	R	1625	9	16	R
1559	2	2	R	1601	3	6	R	1625	11	2	R
1559	2	9	R	1601	3	31	R	1625	11	24	R
1559	2	25	R	1601	4	4	R	1625	11	30	R
1559	3	7	R	1601	12	4	R	1625	12	KU047R-082	R
1559	4	4	R	1601	12	29	R	1625	12	13	R
1559	5	9	R	1602	12	27	R	1625	12	15	R
1559	12	19	R	1603	2	1	R	1625	12	28	R
1559	12	24	R	1603	2	5	R	1626	1	3	R
1560	1	2	R	1603	7	3	R	1626	1	15	B-R
1560	2	1	R	1604	1	19	R	1626	1	25	R
1560	3	23	R	1604	2	27	R	1626	1	26	R
1560	4	17	R	1604	12	15	R	1626	1	27	R
1560	6	24	R	1605	1	17	R	1626	3	6	R
1560	6	26	R	1605	2	10	R	1626	3	18	R
1560	6	29	R	1605	3	8	R	1626	3	19	R
1561	3	14	R	1605	3	11	R	1626	3	20	R
1561	4	7	R	1605	3	21	R	1626	3	21	R
1561	4	14	R	1606	1	30	R	1626	3	22	R
1561	11	8	R	1606	4	7	R	1626	3	23	R
1562	5	7	R	1608	4	2	R	1626	3	25	R
1562	6	7	R	1608	4	3	R	1626	3	29	R
1563	2	23	R	1610	3	25	R	1626	3	31	R
1563	3	16	R	1610	4	10	R	1626	4	2	R
1563	4	18	R	1611	3	10	R	1626	4	3	R

(Continued on next page)

TABLE II

(Continued)

Year	Month	Day	Color	Year	Month	Day	Color	Year	Month	Day	Color
1563	4	22	R	1611	11	29	R	1626	4	18	R
1563	4	26	R	1613	4	16	R	1626	4	20	R
1563	6	17	R	1617	2	6	R	1626	4	22	R
1563	6	26	Y-W	1617	3	4	R	1626	4	28	R
1563	7	24	B-BK	1617	3	21	R	1626	6	7	B-W
1564	2	8	R	1618	11	17	R	1626	6	16	R
1564	2	12	R	1618	12	14	R	1626	6	24	R
1564	3	11	R	1619	1	4	B-W	1626	11	10	B-R
1564	4	15	R	1619	1	5	R	1626	12	7	R
1566	2	23	R	1619	1	7	R	1626	12	11	R
1566	3	26	R	1623	3	28	B-R	1626	12	16	R
1566	7	11	R	1624	2	21	R	1626	12	17	R
1566	8	10	R	1624	2	25	R	1626	12	18	R
1626	12	19	R	1669	10	4	B-R	1721	4	17	R
1626	12	26	R	1675	1	31	Rainbow	1721	4	18	R
1627	1	16	R	1679	11	20	R	1721	4	20	R
1627	1	24	B-W	1679	12	2	R	1721	7	21	R
1627	2	1	R	1680	1	28	R	1721	7	22	R
1627	2	4	R	1681	6	15	R	1722	7	17	R
1627	2	9	R	1681	6	17	R	1722	11	9	R
1628	4	4	R	1681	12	19	R	1723	7	8	R
1628	4	5	R	1682	3	12	R	1723	11	1	R-B
1629	9	4	B-R	1682	3	25	R	1726	5	22	R
1633	1	14	R	1682	4	1	R	1726	7	1	R
1633	1	16	R	1682	4	3	R	1726	7	6	R
1633	2	9	R	1682	4	10	R	1727	4	12	R
1638	3	18	R	1693	4	30	R	1727	4	19	R
1638	11	2	R	1703	11	5	R	1728	4	16	R
1638	11	12	R	1703	11	16	R	1730	3	18	R
1639	4	8	R	1712	9	4	R	1730	5	21	R
1648	11	16	R	1713	11	19	R	1731	7	13	R
1650	8	1	B-R	1717	3	17	R	1747	4	10	R
1660	1	13	R	1717	4	30	R	1753	6	6	R
1660	1	15	R	1720	12	22	R	1756	5	28	R
1660	1	24	R	1721	3	17	R	1779	6	10	R
1660	1	31	R	1721	3	29	R				

We described only the date and color of auroral records in this table, because of page limits.

*Auroral color: R(red), B(blue), W(white), Y(yellow), V(violet), BK(black), R-W(red and white), Y-R(yellow and red) etc.

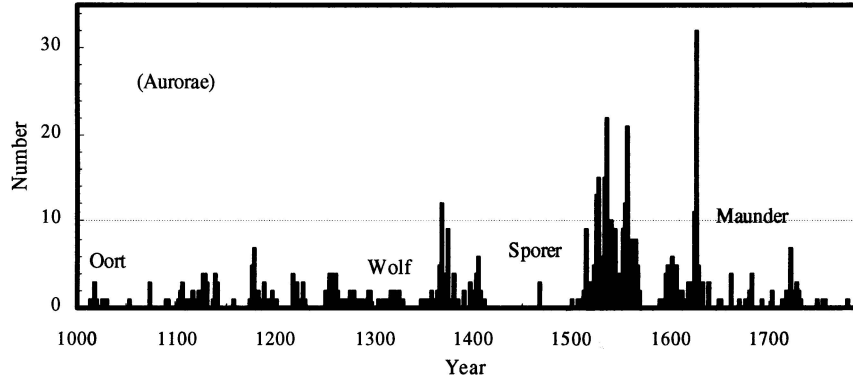


Figure 1. Distribution of auroral records of 11th–18th centuries.

4. Power-Spectrum Analysis

Sunspot and auroral data in historical documents are not homogeneous. Therefore, we have also examined background data (see Figure 2) such as solar and lunar eclipses, and planetary motions etc. in historical records, to estimate the astronomical observation frequency. To estimate the periods of the solar activity, we used a calculating method of the one-dimensional power spectrum from inhomogeneous data. The method of this analysis is derived by the following.

Let $\delta(t)$ denote the time dependent number density field of observational records, t time and ω_j frequency weight. Here, the Fourier transform considering frequency weight is given by

$$\delta_k = \frac{1}{\sum \omega_j} \sum_{j \in T} \omega_j e^{i k t_j}, \quad (1)$$

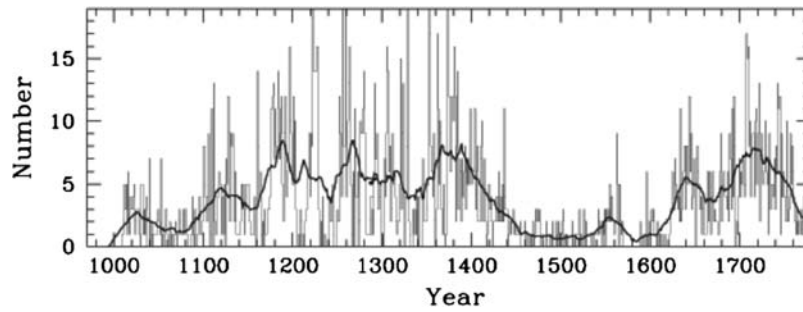


Figure 2. Distribution of background data of 11th–18th centuries.

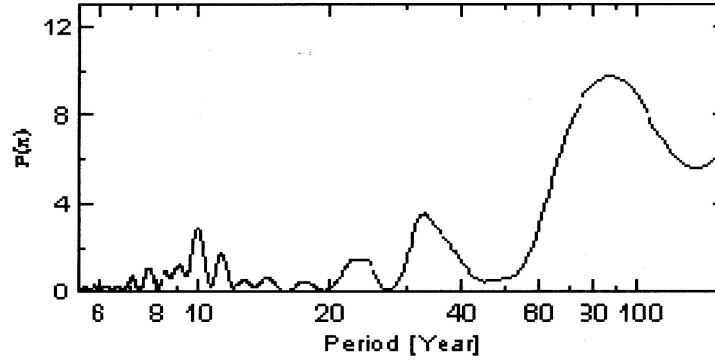


Figure 3. Power spectrum of auroral records during 1397–1799.

where $\kappa (= 2\pi/t)$ is the period and T is the total year of data. Then, the total power spectrum $p(\kappa)$ for the periods is expressed by

$$p(\kappa) = \frac{\left(\langle |\delta_\kappa|^2 \rangle - \frac{\sum \omega_j^2}{(\sum \omega_j)^2} \right)}{\sum_\kappa |\omega_\kappa|^2}, \quad (2)$$

where ω_κ is the Fourier transform during the given period.

Yang, Park, and Park (1998) reported solar cycles in the sunspot and auroral records of the Koryo Dynasty (918–1391). They have found 10.5 year and 97 year cycles for sunspot records, and 10 year and 87 year cycles for auroral records, respectively. In this work, therefore, we have examined the solar cycle in records of the Choson Dynasty. Using 542 auroral observations, we have found 11.2 year and 88.4 year cycles during 1397–1799 (see Figure 3). Unfortunately, we can not obtain the sunspot activity cycle of the Choson period because of insufficient records. That is, auroral activity cycles in each Dynasty show 10 year and 11.2 year in short-term and 87 year and 88.4 year in long-term period, respectively.

Related with our result of a 88.4 year cycle, we note that Feynman (1988) also found a strong line at 88.4 ± 0.7 years verifying the existence of the Gleissberg cycle from the power spectrum analysis of historical auroral data. In the meantime, it is interesting that the 87-year cycle is confirmed by ^{14}C anomaly data during the last 10 000 years (Stuiver and Braziunas, 1988).

5. Discussion

In this work, we have collected Korean historical data from the 11th to 18th centuries. From the distribution of these collected records, we know that high-intensity solar activities strongly existed between the Spörer and Maunder Minima, in spite of the period of low background data. This means that such an activity is only

obtainable from the examination of the long-term solar variation. It is considered that strong solar activities occur before a great deep minimum such as the Maunder Minimum.

The most pronounced feature of solar activity is the 11-year cycle, also called the Schwabe cycle (Usoskin and Mursula, 2003). The 22-year cycle is generally accepted to have exactly twice the period of the 11-year cycle, and corresponds to the switch in polarity of magnetic poles of the Sun (Feynman and Gabrial, 1990). The 88-year Gleissberg cycle of auroral activity is also well established (Gleissberg, 1955; Siscoe, 1980; Feynman, 1988). Moreover, the phase shift of the 88-yr cycle has been also discussed (Kopecký, 1979; Feynman and Gabrial, 1990). Our results derived from the auroral records during 1397–1799, an 11.2 year cycle and an 88.4 year cycle in short- and long-term periods are good evidences of the 11-year Schwabe cycle and the 88-year Gleissberg cycle.

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